



Gabriel Guimarães Carvalho

Research Scientist · Computer Vision · Machine Learning · Ph.D. Physics

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About me

Hi, I'm **Gabriel**. I am a Brazilian mathematical physicist who enjoys transforming mathematics, scientific computing, computer vision, and artificial intelligence into intuitive tools to understand visual data. I bring international experience, having completed my PhD in Italy and now being based in Belgium, where my career gained momentum. I am deeply interested in the future of computing, particularly quantum technologies, as well as the evolving landscape of generative artificial intelligence.

Education

Università di Roma "La Sapienza"

2014 – 2017

Ph.D. in Physics and Relativistic Astrophysics

Roma, IT

Thesis: A walk through mathematical physics with disformal transformations and scalar field self-force effects on a particle orbiting a Reissner-Nordström black hole

Universidade Federal de Pernambuco (UFPE)

2011 – 2013

M.Sc. in Mathematics

Recife, BR

Dissertation (translated): The Ricci Flow and Hamilton's Theorem

Universidade Federal de Pernambuco (UFPE)

2008 – 2010

B.Sc. in Mathematics

Recife, BR

Experience

Research Scientist

Jun. 2024 – Present

imec

Antwerpen, BE

Deep Learning · Computer Vision · Modeling and algorithms · R&D in several domains

Senior Research Scientist

Jan. 2024 – May 2024

IDLab – UAntwerpen – imec

Antwerpen, BE

Deep Learning · Computer Vision · R&D

Computer Vision Data Scientist/Engineer

Jul. 2021 – Nov. 2023

SpaceTime Labs & Moray

São Paulo, BR (Remote)

Computer Vision · Image Processing · Data Preprocessing/Analysis · Hyperspectral Imaging · GIS

Ph.D. Researcher

Sep. 2019 – Jun. 2021

Projeto Cln – Motorola

Recife, BR

Computational Photography · Deep Learning · Data Science/Analysis · Image Processing · Image QA

Postdoctoral Researcher

2017 – 2018

Universidade Federal Rural de Pernambuco (UFPE)

Recife, BR

Alternative gravitational theories and geometric methods in Physics

Lecturer

Nov. 2017 – Jul. 2019

Universidade Federal de Pernambuco (UFPE)

Recife, BR

Calculus I, II, III · Analytic Geometry · Linear Algebra · Complex Variables

Lecturer

Apr. 2012 – Jan. 2014

Universidade Federal de Pernambuco (UFPE)

Recife, BR

Calculus I, II · Analytic Geometry

Skills

ML/CV: PyTorch, PyTorch Lightning, OpenCV, Scikit-Learn, Scikit-Image, NumPy, SciPy, Pandas

Programming: Proficient: Python. Familiar: Go, Matlab, C++, Java

Geospatial: QGIS, GeoPandas, Rasterio

DevOps: Git, Docker, AWS. Familiar: MLFlow, DVC, Apptainer

OS: Linux, macOS, Windows

Spoken: Portuguese (native), English (C1/C2), Italian (B2/C1), Norwegian (A1/A2)

Certifications

DevOps & Software Engineering (IBM) · Deep Learning (DeepLearning.AI) · Math for ML (Imperial College) · ML (Stanford)

Advanced CV with TensorFlow (DeepLearning.AI) · Introduction to Quantum Information (KAIST) · Data Engineering (DeepLearning.AI & AWS)



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